

California Grid Operational Risk Overview

Portfolio project for PG&E Data Visualization Engineer Intern

Grid Reliability + Safety Focus

Project Goal

Build a dashboard-style workflow that turns raw California balancing authority demand data into clear visual summaries.

Core focus:

- Grid reliability
- Operational visibility
- Risk-informed decision making
- Culture of safety

What I Built

Jupyter Notebook

- data cleaning
- UTC-to-Pacific time conversion
- California-only scope validation
- Stress Index engineering

Tableau Dashboard

- demand monitoring
- adequacy view
- balancing authority comparison

Key Skills

- Python data preparation
- Time-series visualization
- Dashboard design
- Scope validation and debugging
- Chart readability improvement
- Decision-support storytelling



Interactive Tableau dashboard for reliability and review monitoring

Key Insight

The first chart looked misleading until the dataset was validated and filtered to California balancing authorities only.

Using a Stress Index relative to each authority's peak demand helped surface hours that may deserve additional review.

Tools Used

Python

pandas
Plotly
Jupyter Notebook
Tableau Public

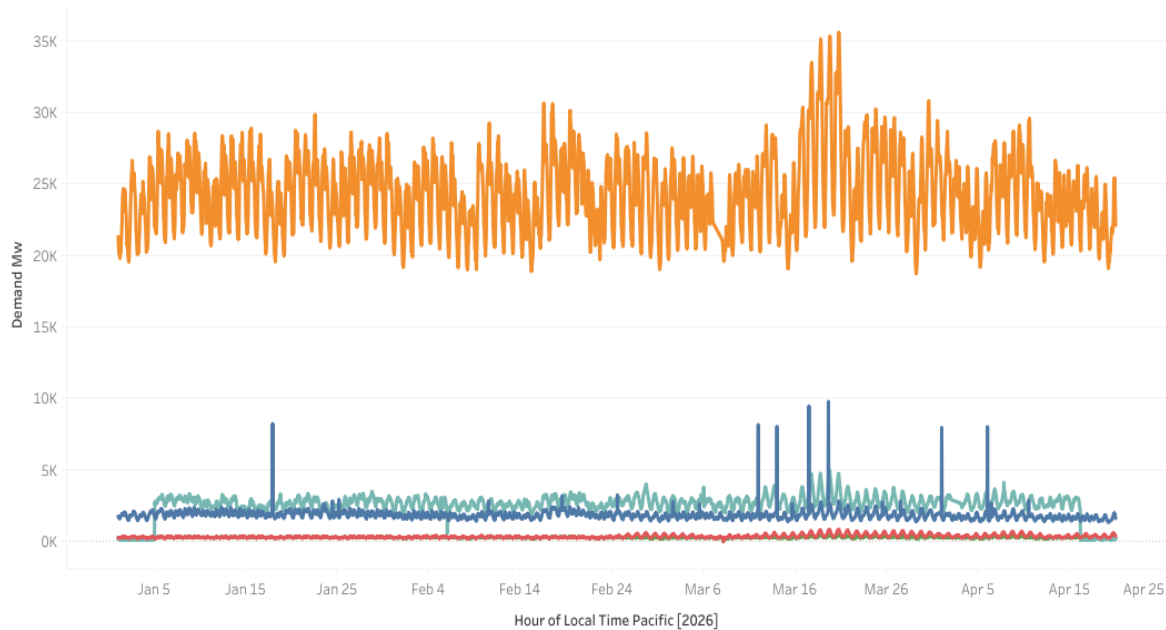
Business Value

- improves visibility into demand conditions
- organizes review-priority hours
- supports clearer interpretation of operational patterns
- demonstrates safety-focused monitoring through visualization

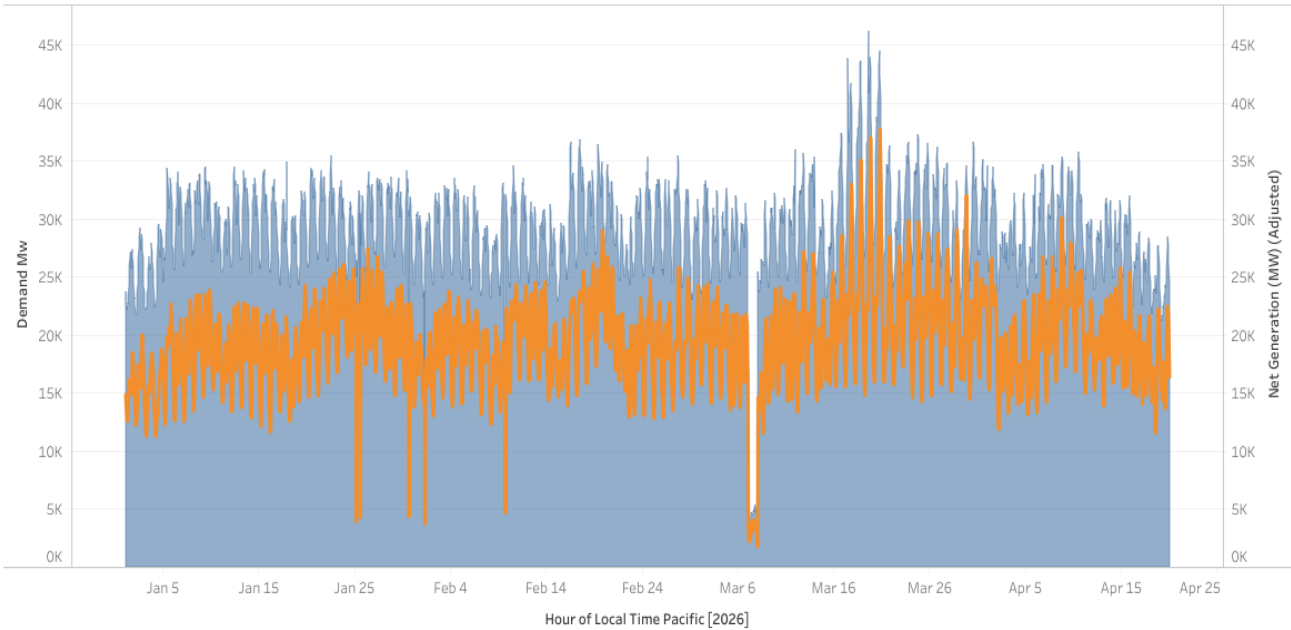
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Executive Summary: This dashboard monitors CA grid reliability using a custom **Stress Index**. By identifying periods where demand approaches peak capacity (90%+), we highlight critical operational risks for March 2026.

Hourly Demand Pulse



Adequacy View



Risk Leaderboard



Review Queue

Review Prio..	Balancing A..	Year of Loc..		
High Review Priority	CISO	2026		2,371
	TIDC	2026		1,806
	IID	2026		1,516
	LDWP	2026		1,219
	BANC	2026		197
Medium Review Priority	CISO	2026		36,789
	TIDC	2026		13,714
	IID	2026		6,374
	LDWP	2026		2,978
	BANC	2026		413